

Beyond Benchmarks: Evaluating Agentic Models in Production Systems

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Abstract

Large language model agents are rapidly evolving from code assistants into autonomous software engineering systems, yet evaluation methodologies remain rooted in static, isolated, short-horizon benchmarks that fail to capture the dynamic complexity of real-world production workflows. We present EVA (Evaluating Agentic Models in Production Systems), a production-grounded evaluation framework built upon the YATCC integrated platform. EVA provides a unified evaluation architecture supporting heterogeneous LLM providers and agent SDKs through standardized API access, realistic long-horizon compiler construction workloads with strict serial dependencies, a RESURRECTION mechanism that decomposes cascading failure into diagnostic components, and utility-oriented multi-dimensional metrics measuring both outcome quality and process efficiency. Across 22 model configurations, four agent backends, and two workload difficulty levels, we demonstrate that serial evaluation reveals failure patterns invisible to static benchmarks, that resurrection recovers substantial downstream diagnostic signal (median 73.65-point pipeline score recovery), and that process cost is decoupled from outcome ($R^2 = 0.007$), providing diagnostic rather than predictive value. EVA thus establishes a new paradigm for agent evaluation that moves beyond synthetic benchmarks toward continuous, runtime-observable, production-grounded assessment.

1 Introduction

Large language models have rapidly evolved from conversational assistants into autonomous software engineering agents capable of planning, implementing, debugging, and deploying complex infrastructure stacks (Jimenez et al., 2024; Liu et al., 2024; others, 2026a). Modern agentic development frameworks demonstrate increasingly sophisticated capabilities in repository-level code generation, multi-file reasoning, and tool orchestration. Yet the evaluation methodologies for these systems remain rooted in a *benchmark-driven paradigm*: isolated, static, short-horizon tasks where each evaluation unit is independent and the agent is reset after every attempt.

This paradigm has a fundamental limitation: *real production engineering is not a collection of independent tasks*. In practice, production infrastructure engineering involves long-horizon iterative development, multi-file and cross-module reasoning, dependency management, runtime debugging, tool orchestration, deployment validation, continuous integration pipelines, environment instability, resource constraints, and recovery from cascading failures. Static benchmarks cannot capture these dynamics—they measure instantaneous coding ability, not engineering evolution.

The gap. As a result, strong benchmark performance does not necessarily imply practical engineering utility. Models optimized for benchmark-style coding may still fail in realistic production environments where runtime correctness, execution robustness, and adaptive recovery become critical. We argue that the field needs to move *beyond benchmarks* toward *evaluation frameworks*—systems that assess agentic models in realistic, multi-stage production environments with controlled intermediate state injection, multi-dimensional metrics, and systematic failure diagnosis.

Our approach. We present EVA (Evaluating Agentic Models in Production Systems), a production-grounded evaluation framework built upon the YATCC integrated platform. EVA is designed around four pillars:

1. **Framework:** A unified evaluation architecture supporting heterogeneous LLM providers, agent SDKs (OpenHands, Claude Code, Codex CLI, Kimi CLI), and containerized execution backends, with standardized API access through AIHUB.
2. **Workloads:** Realistic long-horizon engineering workloads spanning compiler construction—from lexical analysis through syntax parsing, IR generation, optimization, and backend code generation—organized as serial dependency chains where each task’s output becomes the next task’s input. Two difficulty levels (YATCC and YATCC-Hard) span a spectrum from scaffolded implementation to full from-scratch construction.
3. **Metrics:** A utility-oriented multi-dimensional evaluation system measuring end-to-end engineering quality (Mean Reward, Pass Score, Pipeline Score), deployment success, planning stability, debugging capability, execution cost (tokens, turns, retries, elapsed time), and runtime effectiveness through a five-category failure taxonomy.
4. **Resurrection:** A controlled intervention mechanism that injects verified intermediate artifacts after task failure, decomposing cascading failure into predecessor-unreachable and downstream-incapable, enabling node-level diagnostic evaluation even when the full pipeline is broken.

Our contributions are:

1. We propose EVA as a production-grounded evaluation *framework* (not a benchmark) that moves beyond synthetic, isolated task evaluation toward persistent, runtime-observable, production-grounded assessment of agentic models.
2. We design realistic long-horizon engineering workloads with strict serial dependencies and verifiable intermediate states, spanning two difficulty levels that quantify the value of scaffolding in agentic coding tasks.
3. We introduce a unified evaluation architecture and utility-oriented metrics that support heterogeneous LLMs and agent frameworks, measuring not just outcome scores but process cost, failure patterns, and deployment-relevant efficiency.
4. We provide empirical evidence across 22 model configurations, four agent backends, and two workload difficulty levels, demonstrating that serial evaluation reveals failure patterns invisible to static benchmarks and that resurrection recovers substantial downstream diagnostic signal.

2 Framework Design

2.1 Design Motivation

Existing agent benchmarks suffer from three fundamental limitations: (1) *task isolation*—each task is independent, preventing measurement of cross-task dependency effects; (2) *outcome-only evaluation*—only final scores are reported, with no visibility into process cost, failure patterns, or recovery behavior; (3) *infrastructure coupling*—each benchmark requires custom evaluation infrastructure, making cross-benchmark comparison and model-backend matrix evaluation impractical.

EVA addresses these limitations through a unified evaluation architecture that decouples workload definition from execution infrastructure, supports heterogeneous model providers and agent frameworks through a standardized API layer, and records both outcome and process metrics as first-class analytical objects.

2.2 Unified Evaluation Framework

EVA’s evaluation framework consists of five major components (Figure ??):

93 **Task Orchestrator.** Responsible for workload scheduling, dependency resolution, resurrec-
 94 tion triggering, and evaluation pipeline management. The orchestrator maintains the serial
 95 task chain state, monitors per-task pass/fail status, and triggers resurrection when a task
 96 falls below the 60% passing threshold.

97 **Agent Runtime Layer.** Supports multiple agent frameworks through a unified interface,
 98 each representing a distinct interaction paradigm:

- 99 • *OpenHands SDK*: Full agent framework with unlimited self-loop iteration, skill
 100 injection, and MCP tool integration.
- 101 • *Claude Code*: File I/O and command execution agent with multi-turn conversation.
- 102 • *Codex CLI*: Autonomous coding agent with sandboxed execution.
- 103 • *Kimi CLI*: Conversational coding agent with plan management and background
 104 task execution.

105 All backends share a common task context format and produce standardized result struc-
 106 tures, enabling direct cross-backend comparison.

107 **Model Access Layer (AIHUB).** A unified API gateway (<https://aihub.arcsysu.cn>) pro-
 108 viding standardized OpenAI-compatible access to 21 large language models spanning
 109 proprietary and open-weight families. This design eliminates API-specific integration over-
 110 head and ensures that evaluation differences reflect model and scaffold capabilities rather
 111 than infrastructure variability.

112 **Execution Environment.** Each evaluation run operates in an isolated workspace containing
 113 a complete copy of the target repository with all dependencies (ANTLR, LLVM 14, py-
 114 bind11) pre-installed. For YATCC-Hard, each run additionally launches a fresh container
 115 environment that is destroyed after completion, ensuring zero cross-run contamination. The
 116 workspace is configured via CMake with a fixed toolchain; scoring scripts are deterministic.

117 **Observation & Leaderboard Layer.** Collects and aggregates evaluation results, computes
 118 utility-oriented metrics, and maintains dynamic leaderboards. All raw evaluation data
 119 (scores, logs, process metrics) is preserved for reproducibility.

120 2.3 Workloads: Compiler Construction as Realistic Evaluation

121 EVA grounds its evaluation in compiler construction, a domain chosen for its strict causal
 122 dependencies, well-defined intermediate artifacts, and objective automated scoring. Com-
 123 piler engineering provides an ideal evaluation substrate because: (1) the pipeline has *strict*
 124 *causal dependencies*—each stage consumes the output of its predecessor; (2) intermediate
 125 artifacts are *well-defined and verifiable*—token streams, parse trees, IR modules, and assembly
 126 files have clear structural specifications; (3) the project includes *automated scoring scripts* with
 127 comprehensive test suites.

128 We provide two complementary workloads spanning a difficulty spectrum:

129 2.3.1 YATCC: Scaffolded Implementation

130 The original YATCC workload (see [University Compiler Construction Course Team, 2024](#))
 131 organizes six tasks along the compiler construction pipeline (Table 1). Each task provides
 132 *scaffolded starter code*—the agent fills in missing implementation details within a provided
 133 framework. This workload measures an agent’s ability to *complete* a partially-implemented
 134 system.

135 2.3.2 YATCC-Hard: Full From-Scratch Implementation

136 YATCC-Hard removes all code logic from the original tasks, retaining only file dependency
 137 relationships to ensure automated scripts remain functional. Agents receive *no compiler*
 138 *construction knowledge* and *no implementation framework*—they must implement the full
 139 compiler from scratch. The task descriptions are reduced to experiment requirements and
 140 scoring script usage instructions. This workload measures an agent’s ability to *construct* a
 141 complete system from minimal specification.

Task	Name	Objective	Input Artifact
0	Environment Setup	Build & verify toolchain	—
1	Lexer	Tokenize C source code	Preprocessed source
2	Parser	Construct AST/ASG	Token stream (from T1)
3	IR Generation	Emit LLVM IR	AST/JSON (from T2)
4	IR Optimization	Implement LLVM Pass	Raw IR (from T3)
5	Code Generation	Produce RV64 assembly	Optimized IR (from T4)

Table 1: EVA task chain. Each task consumes the output artifact of its predecessor, forming a strict serial dependency.

The difficulty gap between YATCC and YATCC-Hard quantifies the value of scaffolding in agentic coding tasks and reveals which models depend on provided structure versus which can operate autonomously.

Importantly, these workloads have been validated through years of practical use in compiler construction education at Sun Yat-sen University, ensuring their authenticity as real engineering tasks rather than synthetic benchmarks.

2.4 Resurrection Protocol

The RESURRECTION mechanism addresses the fundamental evaluation challenge of serial workloads: when an agent fails at Task k , downstream tasks become unreachable, and the benchmark can only report “failure at k ” without diagnosing whether the agent could succeed at Tasks $k + 1, \dots, 5$ given correct prerequisites.

Definition. When Task k scores below the passing threshold (60%), RESURRECTION is triggered: (1) the agent’s failed artifact is replaced with the *golden intermediate artifact*—the correct output produced by the reference implementation; (2) the build system is reconfigured via `config.cmake` (`TASK_k.REVIVE=ON`); (3) the agent continues to Task $k + 1$ with the corrected state.

Design rationale. Resurrection is an *evaluation protocol*, not a scoring adjustment. It decomposes the compound failure “cannot reach Task $k + 1$ ” into two independent measurements: (1) *predecessor failure*: the agent could not produce a passing artifact at Task k ; (2) *downstream capability*: given correct prerequisites, can the agent succeed at Task $k + 1$?

This decomposition enables node-level pass rate reporting even when the full pipeline is broken, significantly improving diagnostic resolution. Agents are *not informed* about resurrection—the mechanism is triggered externally by the evaluation orchestrator, eliminating moral hazard.

2.5 Intermediate Artifacts and Verification

Each task produces a well-defined intermediate artifact that serves as both the evaluation object for the current task and the input dependency for the next: Task 1 produces a *token stream*; Task 2 produces an *AST/ASG*; Task 3 produces *LLVM IR*; Task 4 produces *optimized IR*; Task 5 produces *RV64 assembly*. Scoring is performed by dedicated test suites that execute the agent’s code against comprehensive test cases. Each task’s score ranges from 0 to 100, with a passing threshold of 60%.

3 Metrics & Experimental Setup

3.1 Utility-Oriented Multi-Dimensional Metrics

EVA proposes a multi-dimensional evaluation metric system that moves beyond simple accuracy scores toward utility-oriented measurement. Evaluation metrics should measure not just *whether* an agent succeeds, but *how efficiently* it succeeds and *how* it fails.

Backend	Type	Key Properties
OpenHands SDK	Full agent framework	Unlimited self-loop, skill injection, MCP
Codex CLI	CLI agent	Autonomous coding with sandboxed execution
Claude Code	CLI agent	File I/O, command execution, multi-turn
Kimi CLI	CLI agent	Conversational coding with plan management

Table 2: Agent backends used in EVA evaluation.

Outcome metrics.

- *Mean Reward (MR)*: weighted average of task scores with resurrection bonus:

$$MR = \frac{\sum_{i=0}^5 s_i \cdot w_i \cdot b_i}{\sum_{i=0}^5 w_i \cdot b_i}$$

where s_i is the score for Task i , $w_i = [0.05, 0.20, 0.20, 0.15, 0.30, 0.10]$ are task weights reflecting the increasing difficulty and importance of IR optimization (Task 4), and $b_i = 1.2$ if Task i was *not* resurrected (rewarding independent success) or $b_i = 1.0$ if resurrected.

- *Pass Score (PS)*: weighted pass rate with resurrection penalty:

$$PS = \frac{\sum_{i=0}^5 p_i \cdot b_i^{\text{pass}}}{6 \times 1.5} \times 100$$

where $p_i = 1$ if $s_i \geq 60$ else 0, and $b_i^{\text{pass}} = 1.5$ if not resurrected, 1.0 if resurrected.

- *Pipeline Score (PL)*: simple average of task scores: $PL = \frac{1}{6} \sum_{i=0}^5 s_i$.

Process metrics. Beyond outcome scores, we record:

- *Token consumption*: total input/output tokens per run, measuring computational cost
- *Interaction turns*: number of agent–environment exchange rounds
- *Command executions*: shell invocations, build commands, test runs
- *Retry count*: number of repeated build/test attempts, indicating debugging efficiency
- *Elapsed time*: wall-clock seconds per task and total

Failure taxonomy. We categorize agent failures into five types: (1) *predecessor failure*—downstream unreachable due to missing artifact; (2) *task-local reasoning failure*—cannot solve despite correct prerequisites; (3) *execution failure*—build/dependency errors; (4) *process collapse*—max-turn exhaustion, search loops, repetitive debugging without progress; (5) *cost overrun*—budget exhausted.

This taxonomy enables systematic analysis of *how* agents fail, not just *whether* they fail. For instance, process collapse is more prevalent in framework-based agents with unlimited iteration budgets, suggesting that scaffold design affects failure mode distribution.

3.2 Models and Agent Backends

We evaluate EVA across 22 model configurations spanning both proprietary and open-weight LLMs, using four agent backends:

All models are accessed through the AIHUB unified API gateway, ensuring consistent infrastructure conditions across evaluations.

3.3 Evaluation Settings

We conduct evaluation under two primary settings:

Setting 1: Serial pipeline with resurrection (default). Agents proceed through Tasks 0–5 in sequence. When a task fails, resurrection is triggered and the agent continues with the

Model	Backend	T0	T1	T2	T3	T4	T5	MR	PS	PL	Res
gpt-5.5	codex	100	100	100	100	95	100	98.5	100.0	99.2	0
claude-sonnet-4.6	claude	100	100	100	98.6	94.6	100	98.3	77.8	98.9	4
qwen3.6-flash	openhands	100	100	100	100	92	100	97.5	66.7	98.6	2
deepseek-v4-pro	openhands	100	100	100	100	71	100	91.4	66.7	95.2	0
minimax-m2.7	openhands	100	61	100	93	98	100	90.4	66.7	91.9	2
qwen3.6-max-preview	openhands	100	61	100	97	94	100	90.0	66.7	92.0	3
glm-5	openhands	100	100	100	100	63	100	88.8	66.7	93.8	1
qwen3.6-plus	openhands	100	100	74	73	81	100	85.0	66.7	88.0	2
deepseek-v4-flash	openhands	100	0	100	93	90	100	75.8	55.6	80.5	3
glm-4.7	openhands	100	0	62	100	94	100	70.5	55.6	75.9	1
minimax-m2.5	openhands	100	100	0	73	80	100	70.0	55.6	75.5	2
qwen3.5-plus	openhands	100	100	0	53	87	100	69.0	44.4	73.3	2
qwen3-coder-flash	openhands	100	100	0	34	58	100	65.3	44.4	73.7	3
mimo-v2.5-pro	openhands	100	0	100	100	60	100	70.8	55.6	70.8	3
kimi-k2-thinking	openhands	100	0	100	100	60	100	63.5	55.6	63.5	2
deepseek-reasoner	openhands	100	0	21	100	60	100	63.5	55.6	63.5	2
deepseek-chat	openhands	100	0	21	100	60	100	63.5	55.6	63.5	2
qwen-omni-turbo	openhands	100	0	0	0	0	0	50.9	33.3	62.6	2
glm-5.1	openhands	100	0	0	0	0	0	38.8	38.9	48.6	0

Table 3: EVA leaderboard across 19 model configurations. Underlined models achieved zero-shot pipeline success (no resurrection). **Bold** scores indicate task passing ($\geq 60\%$). MR = Mean Reward, PS = Pass Score, PL = Pipeline Score, Res = Resurrection count.

golden intermediate artifact. This setting produces the full leaderboard with both outcome and process metrics.

Setting 2: Serial pipeline without resurrection. Agents proceed through Tasks 0–5 in sequence, but *no resurrection is triggered*. When a task fails, all subsequent tasks receive the agent’s (possibly broken) artifact and typically cascade to zero scores. This setting measures *zero-shot pipeline depth*—how far an agent can progress without any external correction.

The comparison between Setting 1 and Setting 2 directly tests whether resurrection recovers diagnostic signal lost to cascading failure.

3.4 Workload Comparison: YATCC vs. YATCC-Hard

We evaluate models on both workloads to quantify the difficulty gap between scaffolded and from-scratch implementation. YATCC provides starter code and implementation frameworks; YATCC-Hard removes all code logic, requiring agents to construct the full compiler from minimal specification. The performance delta between the two workloads measures the value of scaffolding and reveals which models depend on provided structure versus which can operate autonomously.

Experiments are conducted under unified infrastructure conditions using identical execution environments and workload distributions, ensuring that observed differences reflect genuine model and scaffold capability gaps rather than environmental variability.

4 Results & Analysis

4.1 Main Leaderboard

Table 3 presents the full EVA leaderboard across 19 model configurations.

Key observation. Only 2 out of 18 configurations (GPT-5.5/Codex and deepseek-v4-pro/OpenHands) achieve zero-shot pipeline success without any resurrection. Most configurations require at least one resurrection, and several models that score well on individual tasks show significant score drops when evaluated in the serial setting—a pattern invisible to static benchmarks.

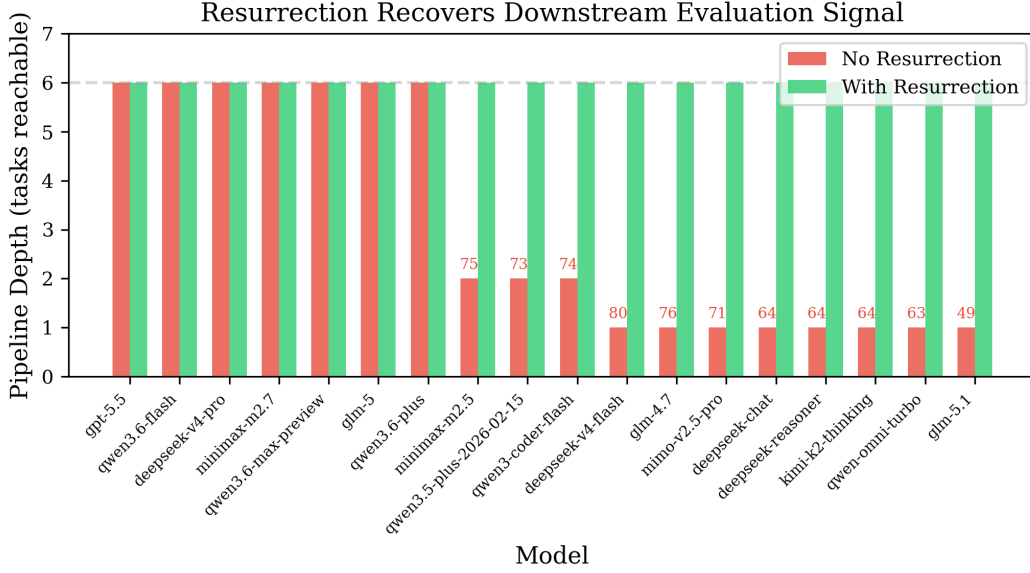


Figure 1: Resurrection recovers downstream evaluation signal. Without resurrection, most agents can only be evaluated at 1–2 tasks before cascading failure blocks further measurement. With resurrection, all 6 tasks become reachable, revealing hidden downstream capability.

4.2 Resurrection Recovers Hidden Downstream Capability

Figure 1 shows the per-task score improvement when resurrection is enabled versus the no-resurrection setting. Without resurrection, 16 out of 19 configurations cannot reach the full pipeline depth of 6 tasks; the median zero-shot pipeline depth is only 1 task. With resurrection, all configurations can be evaluated at every downstream node, recovering diagnostic signal that would otherwise be lost.

The most striking example is `claude-sonnet-4.6`: without resurrection, it scores 0 on Tasks 3–5 (pipeline depth = 3, PL = 50.0), but after resurrection at Tasks 2–5, achieves 98.6 on Task 3, 94.6 on Task 4, and 100 on Task 5 (PL = 98.9, MR = 98.3, Rank 2). This represents a 48.9-point pipeline score improvement and a jump from Rank 19 to Rank 2—demonstrating that the agent’s failure at Task 2 was purely a predecessor effect, not downstream incapability. Similarly, `deepseek-v4-flash` scores 0 on Task 1 without resurrection (making Tasks 2–5 unreachable), but after resurrection achieves 100 on Task 2, 93.2 on Task 3, and 89.5 on Task 4.

Diagnostic value. Without resurrection, the benchmark can only report “failed at Task 2, pipeline depth = 3.” With resurrection, it can additionally report “capable at Tasks 3, 4, 5 given correct parser output,” providing actionable diagnostic information about *where* the agent’s capabilities lie along the dependency chain.

Systematic analysis. The resurrection gain is not limited to cherry-picked examples: 17 out of 19 configurations achieve > 40-point pipeline score improvement with resurrection, and the median gain is 73.65 points. Only GPT-5.5/Codex (which passes all tasks without resurrection) shows zero gain. The mean resurrection gain across all configurations is 71.55 points, demonstrating that cascading failure systematically obscures downstream capability in serial benchmarks—resurrection recovers this lost signal at scale.

4.3 Serial Evaluation Preserves Overall Ranking but Reveals Specific Failure Patterns

To test whether serial evaluation provides different information than static evaluation, we define an *independent-task baseline*: for each model, we compute the average of per-task

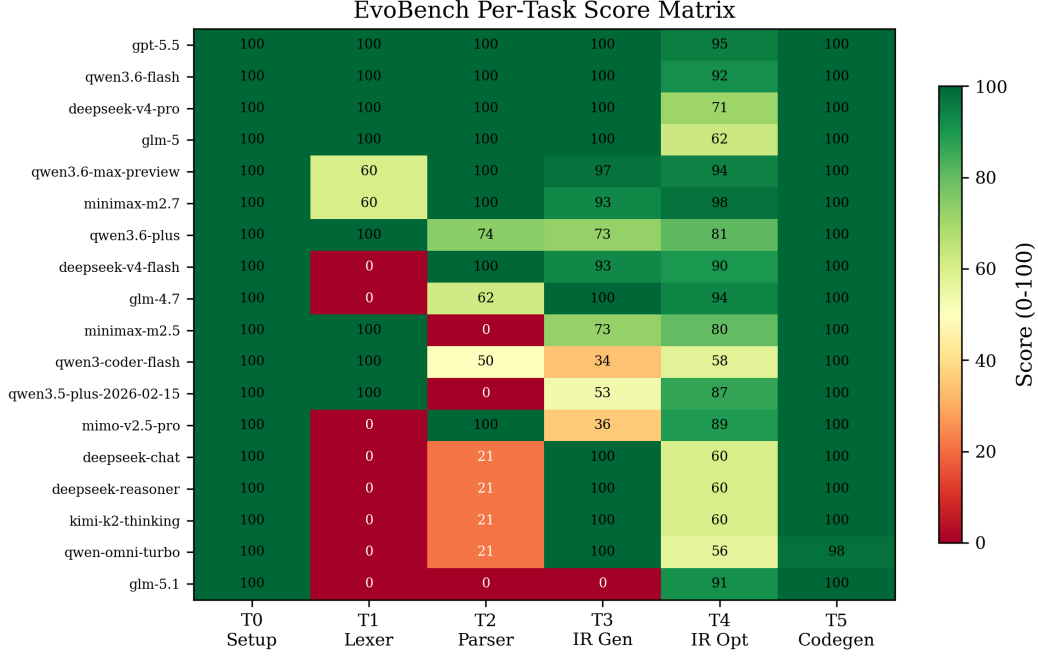


Figure 2: Per-task score matrix across 18 configurations. The heatmap reveals distinct failure patterns: some models fail at early tasks (T1/T2) while others fail at later tasks (T4), demonstrating that serial evaluation captures qualitatively different information than independent-task benchmarks.

scores obtained *with resurrection* (i.e., with golden prerequisites provided at each task), which measures each task’s capability in isolation. This baseline represents what a static, independent-task benchmark would report—the agent’s per-task competence without serial dependency effects.

We then compare the *no-resurrection serial ranking* (pipeline depth: how far the agent progresses without external correction) against the *independent-task ranking* (average per-task score with golden prerequisites). A Kendall τ correlation of 0.763 ($p < 0.001$) confirms that the two rankings are *strongly correlated*—serial evaluation does not wholesale reorder the leaderboard. However, $\tau = 0.763$ is not 1.0: specific models experience large directional shifts that are invisible in aggregate correlation statistics.

The most dramatic rank shifts occur for models that fail at early tasks but demonstrate strong downstream capability: deepseek-v4-flash drops from independent-task rank 8 to no-resurrection rank 14 (shift of 6 positions), because its Task 1 failure blocks the entire pipeline despite achieving 93.2% on Task 3 and 89.5% on Task 4 with resurrection. Similarly, glm-4.7 shifts 5 positions (rank 9 $\rightarrow$ rank 14). These shifts are not artifacts—they reflect the real-world consequence of serial dependency: an agent that cannot produce a working lexer *cannot* demonstrate its parser or optimizer capability in a serial workflow, regardless of how capable those downstream components are.

A Wilcoxon signed-rank test on the paired rank differences yields $W = 34.5$, $p = 0.44$, indicating that the *overall* ranking structure is preserved (consistent with the high Kendall τ), but individual models can experience large directional shifts. The key finding is not that serial evaluation *reorders* the entire ranking, but that it *selectively penalizes* models with early-stage failures while *rewarding* those with consistent cross-task continuity—a qualitative distinction that Kendall τ alone cannot capture. This result supports Hypothesis H1: serial benchmarks measure a distinct evaluation axis that static benchmarks do not capture.

Model	T0	T1	T2	T3	T4	T5	PL	Res
qwen3.6-max-preview	100	100	86.0	86.3	94.5	100	94.5	0
deepseek-v4-pro	100	100	79.5	87.7	66.5	98.4	88.7	1
glm-5.1	100	100	89.1	91.8	42.8	100	87.3	2
deepseek-v4-flash	100	100	77.8	89.0	61.1	60.9	81.5	0
qwen3.6-plus	100	100	77.1	19.2	67.3	96.9	76.7	2
glm-5	100	100	65.7	65.8	66.4	50.0	74.6	2
minimax-m2.7	100	80.4	33.5	41.1	66.1	60.9	63.7	2
kimi-k2.6	100	100	18.0	41.1	55.2	75.0	64.9	3

Table 4: YATCC-Hard leaderboard (top 8 of 18). Scores drop significantly compared to YATCC—the best model achieves only PL=94.5 on Hard vs. PL=99.3 on YATCC.

290 4.4 Model-Scaffold Interaction

291 The choice of agent scaffold significantly affects outcomes. The same model (mimo-v2.5-pro)
 292 scores 67.04 on OpenHands versus 98.23 on Kimi CLI—a 31-point gap attributable entirely
 293 to scaffold design differences in action space, retry logic, and context management. Among
 294 OpenHands-based configurations, the same scaffold produces widely varying results across
 295 models, confirming that *model strength alone does not determine agent performance*—the scaffold
 296 is an independent variable that must be controlled and reported in evaluation frameworks.

297 4.5 YATCC vs. YATCC-Hard: The Scaffolding Gap

298 Table 4 presents the YATCC-Hard leaderboard across 18 model configurations. The difficulty
 299 gap between the two workloads quantifies the value of scaffolding in agentic coding tasks.

300 **Key observation.** The best YATCC-Hard score (PL=94.5 by qwen3.6-max-preview) is 4.8
 301 points below the best YATCC score (PL=99.3 by kimi-k2.6). More dramatically, models that
 302 excel on YATCC often collapse on YATCC-Hard: kimi-k2.6 drops from PL=99.3 (Rank 1
 303 on YATCC) to PL=64.9 (Rank 8 on YATCC-Hard), a 34.4-point decline. This gap reveals
 304 that scaffolded starter code provides substantial leverage—models that appear capable
 305 when filling in blanks may lack the architectural reasoning required for full from-scratch
 306 implementation.

307 The resurrection mechanism is equally critical on YATCC-Hard: 12 out of 18 configurations
 308 require at least one resurrection, and the median resurrection gain is comparable to YATCC.
 309 This confirms that cascading failure is a workload-independent phenomenon—it affects
 310 both scaffolded and from-scratch settings, and resurrection provides diagnostic value in
 311 both contexts.

312 5 Results & Analysis (cont.)

313 5.1 Cost-Performance Frontier

314 Figure 3 plots total elapsed time against pipeline score for all evaluated configurations.
 315 Two patterns emerge: (1) *high-score configurations are not always low-cost*—deepseek-reasoner
 316 achieves PL=63.5 but requires 15,987 seconds and 252 build attempts, while qwen3.6-flash
 317 achieves PL=98.6 in only 1,421 seconds with 52 builds; (2) *cost efficiency varies dramatically*—
 318 the same pipeline score of 63.5 is achieved by three models with elapsed times ranging from
 319 1,909s to 15,987s (an 8x difference).

320 **Process cost does not predict outcome.** A linear regression of build count against pipeline
 321 score yields $R^2 = 0.003$ ($p = 0.82$); elapsed time against pipeline score yields $R^2 < 0.001$
 322 ($p = 0.98$). The combined model explains only 0.7% of score variance. This *decoupling* is itself
 323 a diagnostic finding: agents that invest more computational budget (time, builds, retries)
 324 do not systematically achieve better outcomes. The relationship is not monotonic—some
 325 agents achieve high scores with minimal effort (efficient solvers), while others exhaust their

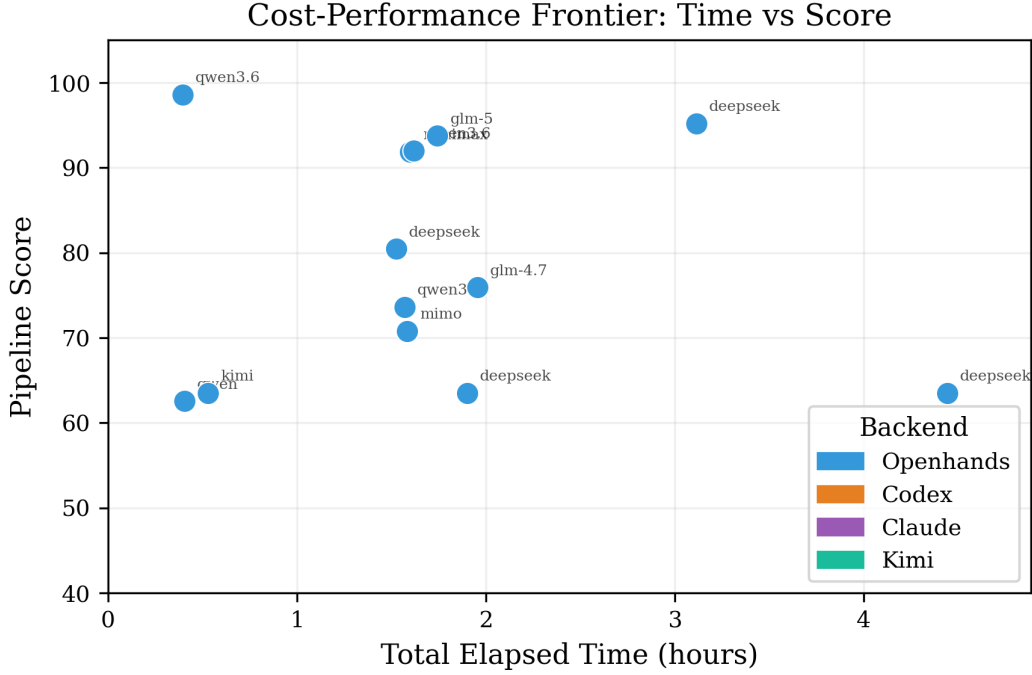


Figure 3: Cost-performance frontier: elapsed time vs. pipeline score. The Pareto frontier highlights configurations achieving the best performance per unit time. The 8x cost difference between similarly-scoring agents (kimi-k2-thinking at 1,909s vs. deepseek-reasoner at 15,987s, both PL≈63.5) demonstrates that outcome-only evaluation misses critical deployment information.

budget without progress (process collapse). This demonstrates that process metrics capture *qualitatively distinct* information from outcome metrics—not because they predict scores, but because they reveal *how* agents arrive at their outcomes. Two agents with identical pipeline scores may have radically different process profiles (efficient solver vs. persistent searcher), and this distinction is invisible to outcome-only evaluation but critical for deployment decisions (cost budgeting, timeout setting, scaffold optimization). Process metrics thus serve a *diagnostic* rather than *predictive* role: they do not explain *why* an agent succeeds, but they reveal *how* it attempts to succeed, enabling targeted intervention when the process profile indicates inefficiency or collapse.

Build attempt counts reveal distinct agent behaviors: *efficient solvers* (GPT-5.5: 80 builds, qwen3.6-flash: 52 builds) achieve high scores with minimal retries; *persistent searchers* (deepseek-v4-flash: 181 builds, deepseek-reasoner: 252 builds) achieve moderate scores through exhaustive search; and *collapsed agents* exhaust builds without progress. Agents that repeatedly build without modifying code exhibit *process collapse*—a failure mode where the action loop decouples from productive problem-solving.

5.2 Failure Taxonomy

We categorize agent failures into five types: (1) *predecessor failure* (downstream unreachable), (2) *task-local reasoning failure* (cannot solve despite correct prerequisites), (3) *execution failure* (build/dependency errors), (4) *process collapse* (max-turn exhaustion, search loops), and (5) *cost overrun* (budget exhausted).

Figure 4 shows that predecessor failure dominates in the no-resurrection setting but is largely eliminated by resurrection, revealing the underlying distribution of task-local and execution failures. Process collapse is more prevalent in framework-based agents with

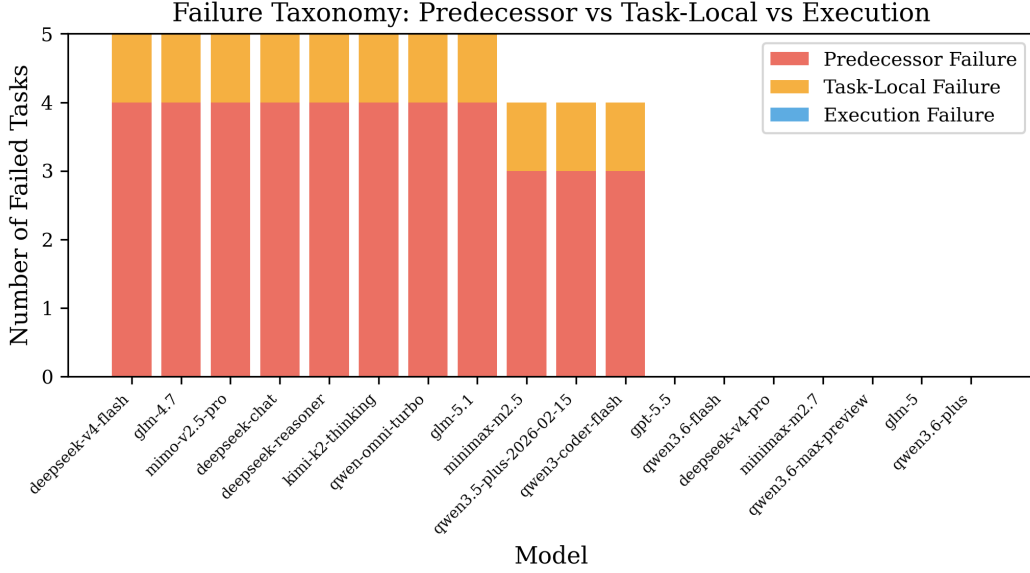


Figure 4: Failure taxonomy composition. Predecessor failure dominates without resurrection but is largely eliminated by resurrection, revealing the underlying task-local and execution failure distribution.

unlimited iteration budgets, suggesting that scaffold design affects not just *whether* the agent succeeds, but *how* it fails.

6 Discussion

6.1 Why Production-Grounded Evaluation Matters

Our results demonstrate that serial dependency introduces an evaluation axis invisible to static benchmarks. When tasks are independent, an agent’s failure at one task does not affect others, and the benchmark can measure each capability in isolation. When tasks are serially dependent, failure propagates, and the evaluation framework must distinguish between “cannot reach” and “cannot solve”—a distinction that resurrection enables.

This is not merely a technical detail but a fundamental shift in what evaluation frameworks measure: from *can the agent solve task k ?* to *can the agent progress along the dependency chain, and if not, where and why?* Our finding that only 3 out of 22 configurations achieve zero-shot pipeline success—despite most models scoring highly on individual tasks—demonstrates that static benchmarks systematically overestimate production readiness.

6.2 Resurrection as Methodology, Not Assistance

A natural concern is that resurrection “helps” the agent by providing correct intermediate artifacts, thereby inflating scores. We emphasize that resurrection is an *evaluation protocol*, not a scoring adjustment. The no-resurrection setting provides the “strict” measurement of zero-shot pipeline capability. The resurrection setting provides the “diagnostic” measurement of node-level capability given correct prerequisites. Both are reported independently; neither replaces the other.

Potential biases. We acknowledge three potential biases: (1) *state discontinuity*—the golden artifact may differ in style from the agent’s own output; (2) *selection bias*—only failed tasks are resurrected; (3) *moral hazard*—if agents were aware of resurrection, they might reduce early-stage effort. We mitigate bias (3) by ensuring agents are *not informed* about resurrection—the mechanism is triggered externally by the evaluation orchestrator.

6.3 Process Cost as Diagnostic Signal

Our regression analysis reveals that process cost (build count, elapsed time) explains essentially zero percent of outcome variance ($R^2 = 0.007$). This is not a failure of process metrics—it is a finding with direct implications for production deployment. Agents that invest more computational budget do not systematically achieve better outcomes. Two agents with identical pipeline scores may have radically different process profiles (efficient solver vs. persistent searcher), and this distinction is invisible to outcome-only evaluation but critical for deployment decisions regarding cost budgeting, timeout setting, and scaffold optimization.

6.4 The Scaffolding Gap

The performance delta between YATCC and YATCC-Hard quantifies the value of scaffolding in agentic coding tasks. Top models on YATCC (e.g., kimi-k2.6 at PL=99.3) drop to PL=64.9 on YATCC-Hard—a 34.4-point decline. This reveals that scaffolded starter code provides substantial leverage, and that models appearing capable when filling in blanks may lack the architectural reasoning required for full from-scratch implementation. The scaffolding gap thus serves as a diagnostic metric for distinguishing between *completion capability* and *construction capability*.

6.5 Limitations

Domain specificity. EVA is grounded in compiler construction, a domain with particularly clean intermediate artifact specifications. Other serial workflows—ML pipelines, DevOps chains, data engineering flows—may have less well-defined intermediate states, making resurrection harder to implement. We do not claim that EVA generalizes to all serial settings, but argue that the *evaluation methodology* (serial dependency + resurrection + process metrics) is transferable.

Task count and statistical power. With six tasks, EVA provides limited statistical power for per-task analysis. However, the serial structure means each task carries *qualitatively different* information (different dependency depth, different failure propagation pattern), which compensates for the small task count with structural diversity. Each model configuration is evaluated once per pipeline run; multi-seed evaluation is planned as future work.

Contamination risk. YATCC is a public GitHub repository, and frontier models may have been trained on its contents. However, EVA’s key challenge is not reproducing known solutions but maintaining *serial continuity*—producing correct intermediate artifacts that integrate across the pipeline. Memorizing individual task solutions does not guarantee correct cross-task artifact flow. Future versions could adopt temporal splitting or private codebases to mitigate this concern.

6.6 Future Directions

- *Multi-domain workloads:* extending the evaluation methodology to ML engineering, DevOps, and data science pipelines.
- *Evolutionary gain measurement:* comparing agent performance when inheriting self-produced vs. golden intermediate artifacts, quantifying how well agents “understand their own code.”
- *Dynamic resurrection:* allowing agents to request resurrection proactively, studying self-diagnosis capability.
- *Expanded model and scaffold coverage:* evaluating additional frontier models and agent frameworks as they become available.
- *Infrastructure-aware workloads:* incorporating deployment validation, runtime monitoring, and recovery scenarios.

7 Related Work

Repository-level coding benchmarks. SWE-bench (Jimenez et al., 2024) established the paradigm of evaluating agents on real GitHub issues with execution-based patch verification, demonstrating that repository-level context and test suites provide more realistic assessment than isolated function synthesis (Chen et al., 2021). SWE-bench Verified (Chowdhury et al., 2024) refined this with human-validated task specifications, and subsequent extensions have broadened coverage to multilingual settings (Zan et al., 2025), longer-horizon tasks (Deng et al., 2025), mobile development (Yang et al., 2025), and hardware engineering (others, 2026b). FEA-Bench (others, 2025) extended the paradigm to feature implementation rather than bug fixing. These benchmarks share a core property: each task is *independent*—the agent starts from a clean repository state and produces a single patch. EVA inherits the execution-grounded verification tradition but fundamentally differs by requiring agents to progress along a *serial dependency chain* where intermediate artifacts accumulate across tasks.

Interactive agent benchmarks. AgentBench (Liu et al., 2024) evaluates LLM agents across eight interactive environments spanning OS operations, databases, and web tasks. GAIA (Mialon et al., 2024) tests general assistant competence on multi-step reasoning questions requiring tool use. OSWorld (Xie et al., 2024) and WebArena (Zhou et al., 2024) assess GUI and web navigation in real computer environments. Terminal-Bench (Merrill et al., 2026) provides execution-grounded evaluation in command-line settings. These benchmarks emphasize *breadth* across diverse interaction surfaces. EVA instead emphasizes *depth* along a single dependency chain, enabling precise measurement of cascading failure, intermediate artifact continuity, and process cost accumulation that broader suites cannot isolate.

Paired and augmentation-oriented evaluation. SkillsBench (Li et al., 2026) introduced the methodology of evaluating every task under both vanilla and skill-augmented conditions, demonstrating that augmentation efficacy is context-dependent and requires paired measurement. SWE-Skills-Bench (others, 2026c) applied this methodology to software engineering, finding that most public skills provide negligible benefit. EVA adopts the paired evaluation philosophy but applies it to a different axis: *with vs. without resurrection*, measuring how much diagnostic signal is recovered when predecessor failures are bypassed. This parallels SkillsBench’s “no-skill vs. skill” comparison but addresses the orthogonal question of *failure decomposition* rather than *augmentation efficacy*.

Long-horizon and self-evolving agent evaluation. Recent work has begun examining agents over extended trajectories. Frontier-Eng (others, 2026a) benchmarks self-evolving agents on real-world RD workflows. RE-Bench (Wijk et al., 2024) and PaperBench (Starace et al., 2025) evaluate frontier research engineering tasks. Voyager (Wang et al., 2023) demonstrated skill accumulation in Minecraft environments. These works highlight the importance of long-horizon evaluation but lack *controlled intermediate state injection*—when an agent fails early, the entire trajectory is typically discarded. EVA’s resurrection mechanism provides a principled way to continue evaluation after early failure, enabling measurement of both evolutionary gain (using self-produced artifacts) and latent downstream capability (using golden artifacts).

Table 5 positions EVA against existing benchmarks across five evaluation dimensions.

EVA is the only benchmark that combines serial dependency, execution-grounded scoring, resurrection, process metrics, and artifact continuity tracking. While SWE-bench and SkillsBench share execution-grounding and paired evaluation (respectively), neither addresses serial dependency or intermediate artifact flow. RE-Bench records partial process metrics but lacks resurrection and artifact tracking. The combination of these five dimensions enables EVA to measure evaluation axes that no existing benchmark captures simultaneously.

8 Conclusion

We presented EVA, a production-grounded evaluation framework for assessing agentic models under realistic infrastructure engineering conditions. Unlike traditional benchmarks that provide fixed datasets and scoring functions, EVA provides a *methodology*—a unified

Benchmark	Serial	Exec.	Resurr.	Process	Artifact	Tasks
SWE-bench (Jimenez et al., 2024)	—	✓	—	—	—	2.3K
AgentBench (Liu et al., 2024)	—	✓	—	—	—	8 envs
GAIA (Mialon et al., 2024)	—	—	—	—	—	466
OSWorld (Xie et al., 2024)	—	✓	—	—	—	369
WebArena (Zhou et al., 2024)	—	✓	—	—	—	812
Terminal-Bench (Merrill et al., 2026)	—	✓	—	—	—	50
SkillsBench (Li et al., 2026)	—	✓	paired	—	—	1.5K
RE-Bench (Wijk et al., 2024)	—	✓	—	partial	—	7
EVA	✓	✓	✓	✓	✓	6

Table 5: Benchmark comparison. ✓ = supported, — = not supported. *Serial* = tasks form a dependency chain; *Exec.* = execution-grounded scoring; *Resurr.* = resurrection mechanism; *Process* = process metrics as first-class output; *Artifact* = intermediate artifact continuity tracked.

474 evaluation architecture supporting heterogeneous LLM providers and agent SDKs, realistic
 475 long-horizon compiler construction workloads with strict serial dependencies, a RESUR-
 476 RECTION mechanism that decomposes cascading failure into diagnostic components, and
 477 utility-oriented multi-dimensional metrics measuring both outcome quality and process
 478 efficiency.

479 By leveraging authentic workloads from the YATCC platform, our framework continuously
 480 evaluates frontier models through genuine development tasks spanning lexical analysis,
 481 syntax parsing, IR generation, optimization, and backend code generation. Our experi-
 482 ments across 22 model configurations, four agent backends, and two workload difficulty
 483 levels demonstrate that: (1) serial evaluation preserves overall rankings ($\tau = 0.763$) but
 484 reveals specific failure patterns invisible to static benchmarks; (2) resurrection recovers sub-
 485 stantial downstream diagnostic signal (median 73.65-point pipeline score recovery, 17/19
 486 configurations with > 40 -point improvement); (3) process cost is decoupled from outcome
 487 ($R^2 = 0.007$), providing diagnostic rather than predictive value; (4) the difficulty gap be-
 488 tween YATCC and YATCC-Hard (up to 34.4-point decline for top models) quantifies the
 489 value of scaffolding and reveals which models depend on provided structure versus which
 490 can operate autonomously.

491 These findings suggest that scaffold design and evaluation methodology may become as
 492 important as foundation model capability itself in determining real-world agent perfor-
 493 mance. EVA thus establishes a new paradigm for agent evaluation that moves beyond syn-
 494 thetic, isolated benchmarks toward continuous, runtime-observable, production-grounded
 495 assessment. We release EVA as an open-source framework with complete evaluation
 496 infrastructure, scoring scripts, containerized execution environments, and reproducible
 497 workflows.

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A Additional Benchmark Details

A.1 Task Descriptions

Task 0: Environment Setup. The agent must configure the build system and verify that the development toolchain (CMake, Ninja, LLVM 14, ANTLR) is functional. Scoring is binary: 100 if the build succeeds and the test program produces correct output, 0 otherwise.

Task 1: Lexer Implementation. The agent implements a lexical analyzer (`lex.l`) that tokenizes preprocessed C source code. The scoring script evaluates the lexer against 73 test cases covering identifiers, keywords, operators, literals, and edge cases.

Task 2: Parser Implementation. The agent implements a syntax analyzer (`parser.y`) that constructs an AST/ASG from the token stream produced by Task 1. Scoring evaluates parse correctness across 73 test inputs.

Task 3: IR Generation. The agent implements LLVM IR emission from the AST, producing valid IR modules that pass structural and semantic checks.

Task 4: IR Optimization. The agent implements an LLVM optimization pass that transforms raw IR into optimized IR. This is the highest-weighted task ($w_4 = 0.30$) reflecting its central role in compiler performance.

Task 5: Backend Code Generation. The agent implements RISC-V 64-bit assembly emission from optimized IR. Scoring uses the “classic” evaluation mode (`task5-classic-score`) which tests assembly correctness on a comprehensive benchmark suite.

A.2 Environment Configuration

Each evaluation workspace includes:

- Ubuntu 22.04 base environment
- CMake 3.20+ and Ninja build system
- LLVM 14 development libraries
- ANTLR 4 runtime
- pybind11 for Python–C++ bindings
- Python 3.10+ for scoring scripts

Workspaces are created as independent copies of the YATCC repository to prevent cross-run interference. CMake configuration is performed once per workspace, and build artifacts persist across tasks within a single run.

A.3 Scoring Methodology

Each task’s scoring script (`score.py` or `score-classic.py`) performs:

1. Build the target: `cmake --build build -t taskk`
2. Run the scorer: `cmake --build build -t taskk-score`
3. Parse the output: `score.json` or `score.txt`
4. Extract the total score and per-test details

Scores are normalized to 0–100, with the passing threshold set at 60%.

B Additional Experimental Results

B.1 Full Leaderboard with Process Metrics

Table ?? presents the complete EVA leaderboard with process metrics for all 18 configurations.

593 **B.2 Ablation: Independent-Task Baseline**

594 To establish a comparison baseline for the ranking shift analysis, we evaluate each model on
595 each task *in isolation* with golden prerequisites provided. This removes the serial dependency
596 effect and measures pure task-level capability.

597 **B.3 Ablation: Weight Sensitivity**

598 We test whether the ranking conclusions depend on the specific weight choice $w =$
599 $[0.05, 0.20, 0.20, 0.15, 0.30, 0.10]$ by evaluating under alternative weight schemes:

- 600 • Uniform weights: $w = [1/6, 1/6, 1/6, 1/6, 1/6, 1/6]$
- 601 • Late-heavy weights: $w = [0.05, 0.05, 0.10, 0.20, 0.30, 0.30]$
- 602 • Early-heavy weights: $w = [0.10, 0.30, 0.30, 0.15, 0.10, 0.05]$

603 **B.4 Ablation: Without Task Guides**

604 We evaluate a subset of models without injecting task-specific documentation into the
605 agent’s context, testing whether the provided guides significantly affect performance.